Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

# Results

01 · GRANGER CAUSALITY

does X predict future Y?

**Table.** Granger tests (both directions)

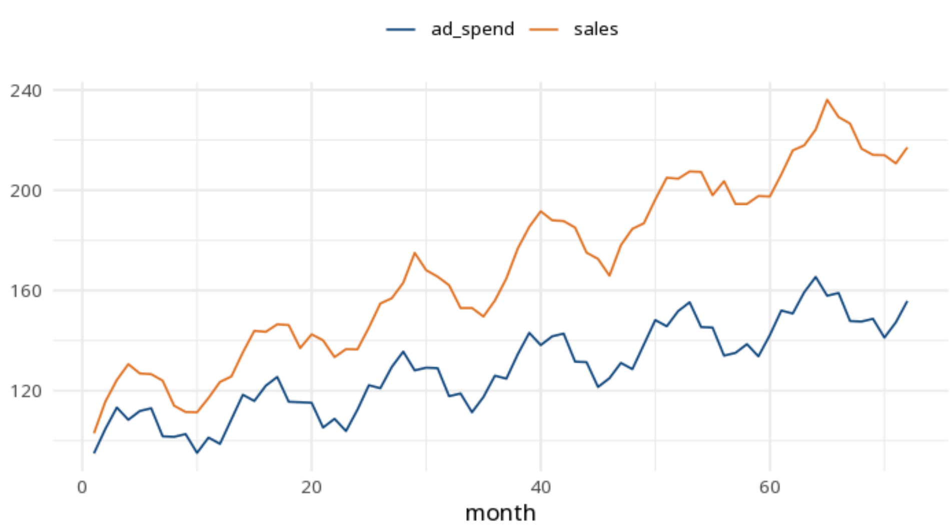
| Direction | F | df | p |
|---|---|---|---|
| X→Y | 47.89 | 4, 63 | <.001 |
| Y→X | 30.62 | 4, 63 | <.001 |

**Table 2.** Lag-order selection (advisory)

| Lag | AIC | BIC | HQ |
|---|---|---|---|
| 1 | 6.78 | 6.98 | 6.86 |
| 2 | 5.80 | 6.12 | 5.93 |
| 3 | 5.36 | 5.81 | 5.54 |
| 4 | 4.96 | 5.55 | 5.19 |

tests predictive precedence, not true causation; both series should be stationary first. Tests use a max lag of 4; select the lag on theory or by AIC/BIC (vars::VARselect) and report it.

**Figure.** Series together




## Understanding the numbers

**Direction.** Which direction this row tests - whether the first series helps predict the second, or vice versa. Here, this row tests X⟶Y.

**F.** The F-statistic testing whether past values of one series improve prediction of the other, beyond the other's own past. Here, $F = 47.89$.

**df.** The degrees of freedom for the F-test (numerator, denominator). Here, df = (4, 63).

**p.** A significant p means past values of X help predict Y beyond Y's own past - predictive precedence, not proof of causation. Here, $p < .001$.

**Lag.** A candidate lag order compared by information criteria - the note above uses a fixed max lag regardless of what this table recommends. Here, AIC is minimized at lag = 4.

**AIC.** An information criterion used to pick the lag order - lower is better relative to the alternatives compared. Here, AIC = 4.96 at the AIC-minimizing lag.

**BIC.** Like AIC but penalizes complexity more heavily; used alongside it to pick the lag order. Here, BIC = 5.55 at the AIC-minimizing lag.

**HQ.** The Hannan-Quinn criterion: another information criterion for comparing candidate lag orders. Here, HQ = 5.19 at the AIC-minimizing lag.

### How to read this test

A significant X⟶Y p means past values of X help predict Y beyond Y's own past — this is predictive precedence, not proof that X causes Y. Check both directions. The result depends entirely on the lag order (this card uses a max lag of 4): choose it on theory or by an information criterion — vars::VARselect reports the AIC- and BIC-minimising lag — and report it, since different lags can flip the conclusion. Make both series stationary first. Method: R lmtest::grangertest (Granger, 1969). Your significance threshold ($\alpha$) is 0.05.

**APA template:** Granger test X→Y: $F(4,63)=47.89$, $p < .001$; Y→X: $F(4,63)=30.62$, $p < .001$ (lag=4).

**Statistical basis:** Granger causality F-test (Granger, C. W. J. (1969). "Investigating causal relations by econometric models and cross-spectral methods." Econometrica, 37(3), 424-438.); Implementation (lmtest::grangertest) (Zeileis A, Hothorn T (2002). "Diagnostic Checking in Regression Relationships." R News, 2(3), 7-10.). Full references in the exported CITATIONS.txt.